STUDY-UNI · UCC MARKING SCHEME

# MA1100
# Introductory Mathematics for Business I

Full Model Answers & Mark Allocation

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**Examination Session:** Summer 2024 · **Paper:** 1

**Module Code:** MA1100 · **Duration:** 1.5 hours

**Instructions to Candidates:** Answer Question 1 and two other questions.

**Prepared by:** Study-Uni

## Question 1

Total: 30 marks

### **(a)** [5 marks]

*€20,000 is invested at a rate of 5% per annum compounded yearly. What is the value of the investment after 25 years?*

## FULL MODEL ANSWER

Apply the compound interest formula $A = P(1 + r)^t$ with $P = €20,000$, $r = 0.05$, $t = 25$:

$$A = 20\{,\}000 \times (1.05)^{25}$$

$$A = €67,727.10 \text{ (to the nearest cent)}$$

### Mark Allocation — (a)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **5** | **Correct formula with correct substitution AND correct final value €67,727.10 (± €0.01).** |
| Good | 3 | Correct formula and substitution, minor rounding/calculator slip in the final figure. |
| Partial | 2 | Correct formula stated with correct values identified (P, r, t) but not correctly evaluated. |
| Minimal | 0 | Incorrect formula (e.g. simple interest) or no coherent attempt. |

### (b) [5 marks]

*Calculate the monthly repayments on a loan of €50,000 borrowed at a rate of 3.6% per annum over 8 years.*

#### FULL MODEL ANSWER

Use the present value annuity formula $V_0 = A_0[1 - (1+i)^{-n}] / i$, and solve for the monthly repayment $A_0$. Here $V_0$ = €50,000, monthly rate $i = 0.036/12 = 0.003$, and $n = 8 \times 12 = 96$ months:

$$A_0 = V_0 \times i / [1 - (1+i)^{-n}] = 50{,}000 \times 0.003 / [1 - (1.003)^{-96}]$$

$$A_0 = €600.20 \text{ per month (to the nearest cent)}$$

**Checknote:** *This is the mirror image of the mortgage-term problems elsewhere on this paper: here n and i are known and $A_0$ is the unknown, so the present value formula must be rearranged rather than solved directly.*

#### Mark Allocation — (b)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **5** | **Correctly identifies $V_0$ = 50,000, i = 0.003, n = 96 AND rearranges the present value annuity formula for $A_0$ AND reaches €600.20.** |
| Good | 3 | Correct formula and correct i, n, with a slip in the final evaluation. |
| Partial | 2 | Correct formula identified but incorrect conversion of the annual rate/term to monthly (e.g. using n = 8 instead of 96). |
| Minimal | 0 | Wrong formula used (e.g. future value formula instead of present value), or no coherent attempt. |

## (c)(i) [2 marks]

*Calculate the Annual Percentage Rate (APR) for an interest rate of 4% compounded monthly, to 4 decimal places.*

### FULL MODEL ANSWER

$APR = (1 + r/m)^m - 1$, with $r = 0.04$, $m = 12$:

$$APR = (1 + 0.04/12)^{12} - 1$$

$$APR = 4.0742\%$$

### Mark Allocation — (c)(i)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **2** | **Correct formula, correct substitution, AND APR = 4.0742% to 4dp.** |
| Minimal | 0 | Incorrect or missing formula. |

## (c)(ii) [1 marks]

*Calculate the Annual Percentage Rate (APR) for an interest rate of 4% compounded continuously, to 4 decimal places.*

### FULL MODEL ANSWER

For continuous compounding, APR = $e^r - 1$, with $r = 0.04$:

$$\text{APR} = e^{0.04} - 1$$

$$\text{APR} = 4.0811\%$$

**Checknote:** *As in every APR-comparison question on this paper: continuous compounding always gives the highest APR for a given nominal rate — a fast self-check between (c)(i) and (c)(ii).*

### Mark Allocation — (c)(ii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **1** | **Correct continuous-compounding formula AND APR = 4.0811% to 4dp.** |
| Minimal | 0 | Uses the discrete-compounding formula instead of $e^r$, or no coherent attempt. |

## **(d)(i)** [2 marks]

*Find the derivative of $f(x) = 4x^3 + 7x + \sqrt{x}$ and write your answer in its simplest form.*

### FULL MODEL ANSWER

Differentiate term by term, writing $\sqrt{x}$ as $x^{1/2}$:

$$f(x) = 4x^3 + 7x + x^{1/2}$$

$$f'(x) = 12x^2 + 7 + 1/(2\sqrt{x})$$

### Mark Allocation — (d)(i)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **2** | **Correct power-rule differentiation of all three terms AND simplified final answer.** |
| Partial | 1 | Correct differentiation of two of the three terms. |
| Minimal | 0 | Incorrect application of the power rule. |

### **(d)(ii)** [2 marks]

*Find the derivative of $f(x) = x(\ln x - 1)$ and write your answer in its simplest form.*

#### FULL MODEL ANSWER

Apply the **product rule**, $u = x$, $v = (\ln x - 1)$, so $u' = 1$, $v' = 1/x$:

$$f'(x) = 1\cdot(\ln x - 1) + x\cdot(1/x) = (\ln x - 1) + 1$$

$$f'(x) = \ln x$$

#### Mark Allocation — (d)(ii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **2** | **Correct product rule setup AND simplifies fully to ln x.** |
| Partial | 1 | Correct product rule setup but does not simplify to ln x. |
| Minimal | 0 | Product rule not applied, or incorrect derivative of ln x. |

## (d)(iii) [2 marks]

*Find the derivative of $f(x) = (x^2 - 1)/(x - 1)$ and write your answer in its simplest form.*

### FULL MODEL ANSWER

Rather than applying the quotient rule directly, notice the numerator factorises: $x^2 - 1 = (x-1)(x+1)$. For $x \neq 1$, the function simplifies first:

$$f(x) = (x-1)(x+1) / (x-1) = x + 1$$

This is now trivial to differentiate:

$$f'(x) = 1$$

**Checknote:** *Spotting that $x^2-1$ factorises as $(x-1)(x+1)$ turns this into a one-line derivative — always check whether a rational function simplifies before reaching for the quotient rule.*

### Mark Allocation — (d)(iii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **2** | **Recognises the difference-of-squares factorisation, simplifies $f(x)$ to $x+1$ AND correctly states $f'(x) = 1$.** |
| Good | 1 | Applies the quotient rule directly without simplifying first and reaches an equivalent correct expression that simplifies to 1. |
| Minimal | 0 | Neither the simplification route nor a correctly applied quotient rule is present. |

### **(d)(iv)** [3 marks]

*Find the derivative of $f(x) = 1/\sqrt{(x+1)}$ and write your answer in its simplest form.*

#### FULL MODEL ANSWER

Write as a power and apply the **chain rule**:

$$f(x) = (x+1)^{-1/2}$$

$$f'(x) = -(1/2)(x+1)^{-3/2} \times 1$$

$$f'(x) = -1 \,/\, [2(x+1)^{3/2}]$$

#### Mark Allocation — (d)(iv)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **3** | **Correctly rewrites as a negative power AND correct chain rule application AND correctly simplified final form.** |
| Good | 2 | Correct chain rule application with an index-law slip in the final simplification. |
| Partial | 1 | Correct power rewritten but chain rule (inner derivative) omitted. |
| Minimal | 0 | No coherent differentiation attempt. |

### **(e)** [8 marks]

*Using implicit differentiation, find dy/dx for $x^2 + y^2 - 6x - 6y + 10 = 0$. Given that $a = (5,1)$ is a point on the graph, find the slope of the tangent line at the point a. Hence, find the equation of the tangent line at the point a.*

**FULL MODEL ANSWER**

Differentiate every term with respect to *x*, treating *y* as a function of *x* ($y^2$ needs the **chain rule**):

$$2x + 2y(dy/dx) - 6 - 6(dy/dx) = 0$$

Collect the dy/dx terms on one side:

$$(dy/dx)(2y - 6) = 6 - 2x$$

$$dy/dx = (6 - 2x) / (2y - 6) = (3 - x) / (y - 3)$$

**Slope at a(5,1):** substitute $x = 5$, $y = 1$:

$$dy/dx = (3 - 5) / (1 - 3) = (-2) / (-2)$$

$$\text{slope} = 1$$

**Tangent line at a(5,1):** use point-slope form $y - y_1 = m(x - x_1)$:

$$y - 1 = 1(x - 5)$$

$$y = x - 4$$

**Checknote:** *This equation describes a circle — $(x-3)^2 + (y-3)^2 = 8$ — so the tangent line at (5,1) can be sanity-checked by confirming it is perpendicular to the radius from the centre (3,3) to (5,1), which has slope −1, matching a tangent slope of 1.*

### Mark Allocation — (e)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **8** | **Correct implicit differentiation of all terms (chain rule on $y^2$) AND correct general dy/dx AND correct slope at a (= 1) AND correct tangent line equation.** |
| Good | 6 | Correct dy/dx and correct slope at a, but an error in forming or simplifying the final tangent line equation. |
| Partial | 4 | Correct differentiation of each term individually but an error collecting dy/dx terms, so the general dy/dx expression is wrong — slope and tangent line follow through correctly from the student's own (incorrect) expression. |
| Minimal | 1 | Chain rule omitted on $y^2$ (treated as if y were a constant). |

**Question 1 — Total** 30/30 marks confirmed

## Question 2

Total: 25 marks

### **(a)** [5 marks]

*The supply function is $P_s = (1/4)Q + 3/2$ and the demand function is $P_d = 15 - (1/8)Q$. If P is on the vertical axis and Q is on the horizontal axis, write down the slope of the supply function and the slope of the demand function.*

#### FULL MODEL ANSWER

Both functions are already written as P = (slope)·Q + intercept, so the slopes read off directly:

| Function | Equation | Slope (dP/dQ) |
|---|---|---|
| Supply | $P_s = (1/4)Q + 3/2$ | **1/4** |
| Demand | $P_d = 15 - (1/8)Q$ | **−1/8** |

#### Mark Allocation — (a)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **5** | **Both slopes correctly identified: supply = 1/4, demand = −1/8, with correct sign on the demand slope.** |
| Partial | 3 | One slope correct, or both magnitudes correct but the negative sign on demand omitted. |
| Minimal | 0 | Slopes not correctly extracted from the given equations. |

### **(b)** [6 marks]

*Determine the equilibrium price and quantity.*

## FULL MODEL ANSWER

At equilibrium, $P_s = P_d$:

$$(1/4)Q + 3/2 = 15 - (1/8)Q$$

$$(3/8)Q = 27/2 \;\Rightarrow\; Q = 36$$

Substitute back into either equation to find P:

$$P = (1/4)(36) + 3/2 = 9 + 1.5$$

**EQUILIBRIUM**

$$Q^* = 36,\;\; P^* = €10.50$$

### Mark Allocation — (b)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **6** | **Correctly sets $P_s = P_d$, solves for Q = 36 AND P = €10.50, with both values stated clearly.** |
| Good | 4 | Correct method and correct Q, with an arithmetic slip carried into the final P value (or vice versa). |
| Partial | 2 | Correct equation set up ($P_s = P_d$) but not solved through to numeric values. |
| Minimal | 0 | Incorrect equilibrium condition used. |

### **(c)** [8 marks]

> *If a tax of €4.50 is added on the sale of each item, what will the new equilibrium price and quantity be?*

#### FULL MODEL ANSWER

A per-unit tax raises the price sellers must receive to be willing to supply any given quantity — it shifts the supply curve **up** by the amount of the tax:

$$P_{s(\text{new})} = (1/4)Q + 3/2 + 4.5 = (1/4)Q + 6$$

Set the new supply curve equal to the (unchanged) demand curve:

$$(1/4)Q + 6 = 15 - (1/8)Q$$

$$(3/8)Q = 9 \;\Rightarrow\; Q = 24$$

Substitute into the demand curve to get the price consumers pay:

$$P = 15 - (1/8)(24) = 15 - 3 = 12$$

$$\text{New } Q^* = 24,\;\; \text{new price paid by consumers} = €12.00$$

The price producers actually *keep* per unit, net of the tax, is this consumer price minus the €4.50 tax: €12.00 − €4.50 = **€7.50** — this figure is needed for part (d).

> **Checknote:** *A tax pushes the supply curve up (sellers need more per unit to break even after remitting the tax) — the opposite direction to a subsidy, which pushes it down.*

### Mark Allocation — (c)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **8** | **Correctly shifts the supply curve up by €4.50 AND solves the new equilibrium AND correctly reports Q = 24, consumer price = €12.00 (with producer's net price identified as €7.50).** |
| Good | 6 | Correct direction of shift and correct new equilibrium Q and consumer price, but the distinction between consumer price and producer's net price not drawn out. |
| Partial | 3 | Correctly identifies that supply shifts by €4.50 but shifts it the wrong direction (down instead of up), carrying through consistently. |
| Minimal | 0 | Tax applied to the demand curve instead of supply, or no shift applied at all. |

### **(d)** [6 marks]

> *In what ratio will the tax be shared between the producer and the consumer?*

## FULL MODEL ANSWER

Compare each side's price before and after the tax (using the original equilibrium price €10.50 from part (b) as the baseline):

| Party | Price before | Price after | Burden |
|---|---|---|---|
| Consumer pays | €10.50 | €12.00 | **€1.50** |
| Producer receives (net) | €10.50 | €7.50 | **€3.00** |
| **Total (must equal the €4.50 tax)** | | | **€4.50** |

Expressing the burdens as a ratio, producer : consumer:

$$3.00 : 1.50 \;\equiv\; 2 : 1$$

The producer bears twice the burden of the consumer here. This follows from the relative slopes in part (a): supply is flatter (more elastic, |slope| = 1/4) than demand (less elastic, |slope| = 1/8) — the flatter (more elastic) side of the market bears the *larger* share of a tax, since it is less able to simply absorb the price change by adjusting quantity as freely as the steeper side.

> **Checknote:** *Note the ratio direction is reversed compared with a market where demand is the flatter curve — always re-derive from the actual slopes rather than assuming which side 'always' bears more.*

### Mark Allocation — (d)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **6** | **Correctly computes both the consumer price rise (€1.50) and producer price fall (€3.00), AND confirms they sum to €4.50, AND states the ratio 2:1 (producer:consumer).** |
| Good | 4 | Correct ratio 2:1 obtained, without explicit reference to the sum-to-€4.50 check or the elasticity explanation. |
| Partial | 2 | One of the two burden amounts (consumer rise or producer fall) correctly computed, but the ratio not correctly formed. |
| Minimal | 0 | No coherent attempt to split the €4.50 tax between the two parties. |

**Question 2 — Total** 25/25 marks confirmed

## Question 3

Total: 25 marks

## **(a)** [8 marks]

*A couple requires €50,000 in 18 years' time, when their child starts college. They decide to invest €150 every month for 18 years at 4.8% per annum compounded monthly. What will be the value of the investment at the end of the 18 years?*

### FULL MODEL ANSWER

Use the future value annuity formula V_n = $A_0[(1+i)^n - 1] / i$, with the monthly deposit $A_0$ = €150, monthly rate i = 0.048/12 = 0.004, and n = 18 × 12 = 216 months:

$$V_{216} = 150 \times [(1.004)^{216} - 1] / 0.004$$

$$V_{216} = €51{,}320.50 \text{ (to the nearest cent)}$$

### Mark Allocation — (a)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **8** | **Correctly identifies $A_0$ = 150, i = 0.004, n = 216 AND applies the future value annuity formula correctly AND reaches €51,320.50.** |
| Good | 6 | Correct formula and correct i, n, but a slip in the final evaluation. |
| Partial | 4 | Correct formula identified but incorrect conversion of the annual rate/term to monthly (e.g. using n = 18 instead of 216). |
| Minimal | 1 | Wrong formula used (e.g. present value formula, or simple compound interest with no annuity structure). |

### **(b)** [8 marks]

*The couple win a local lottery and decide they would prefer to invest a lump sum now instead of saving monthly. How much would they need to invest in the same account to have €50,000 in 18 years' time? How much would they save by doing this?*

#### FULL MODEL ANSWER

**Lump sum required:** discount the target €50,000 back to today at the same monthly rate i = 0.004 over n = 216 months:

$$V_0 = 50\{,\}000 / (1.004)^{216}$$

$$V_0 = €21{,}109.99 \text{ (to the nearest cent)}$$

**Savings:** compare this lump sum to the total cash the couple would otherwise pay in via monthly deposits (150 × 216 months):

$$\text{Total monthly contributions} = 150 \times 216 = €32{,}400.00$$

$$\text{Savings} = 32\{,\}400.00 - 21\{,\}109.99$$

$$\text{Savings} = €11{,}290.01$$

> **Checknote:** *The 'saving' here is about total cash paid out of pocket, not investment return — it's the gap between what the couple hands over as monthly deposits versus as a single lump sum, since the lump sum benefits from a full 18 years of compounding on the whole amount immediately.*

### Mark Allocation — (b)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **8** | **Correctly discounts €50,000 back to a lump sum of €21,109.99 AND correctly compares it to total monthly contributions of €32,400 AND reaches savings of €11,290.01.** |
| Good | 6 | Correct lump sum value found, but the savings comparison uses the wrong benchmark (e.g. compares to the €50,000 target rather than total contributions paid in). |
| Partial | 4 | Correct present-value formula and substitution attempted, but not correctly evaluated. |
| Minimal | 1 | Future value formula used instead of present value, or no coherent attempt. |

### **(c)** [9 marks]

*The monthly repayment on a mortgage of €400,000 is €1,410. If the interest rate is 2.4% per annum compounded monthly, how many years does it take to pay off the mortgage?*

#### FULL MODEL ANSWER

Use the present value annuity formula $V_0 = A_0[1 - (1+i)^{-n}] / i$, with $V_0$ = €400,000, $A_0$ = €1,410, i = 0.024/12 = 0.002. Solve for n:

$$400{,}000 = 1{,}410 \times [1 - (1.002)^{-n}] / 0.002$$

$$1 - (1.002)^{-n} = 400{,}000 \times 0.002 / 1{,}410 = 0.56738$$

$$(1.002)^{-n} = 0.43262$$

Take logs of both sides and solve for n:

$$-n \cdot \ln(1.002) = \ln(0.43262) \;\Rightarrow\; n = 419.36 \text{ months}$$

$$n \approx 419 \text{ months} \approx 34.9 \text{ years}$$

**Checknote:** *Unlike some mortgage-term questions on other sittings, this one does not resolve to a perfectly round number of years — 34.9 years is the correct answer here, and should not be forced to round to 35 without showing the underlying figure.*

### Mark Allocation — (c)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | 9 | **Correctly sets up the present value annuity formula with the given values AND correctly isolates $(1.002)^{-n}$ AND solves via logs to n ≈ 35 years.** |
| Good | 7 | Correct set-up and correct isolation of the exponential term, with a slip in the final log-solving step. |
| Partial | 5 | Correct formula identified and correctly substituted, but not successfully rearranged to isolate the exponential term. |
| Minimal | 2 | Uses the future value formula instead of present value, or does not attempt to solve for n via logarithms. |

**Question 3 — Total** 25/25 marks confirmed

## Question 4

Total: 25 marks

## 4(a)(i) [5 marks]

*A supply function is given by $Q = 40 + 0.1P^2$. Calculate the elasticity equation as a function of P.*

### FULL MODEL ANSWER

Price elasticity of supply is $E = (dQ/dP) \times (P/Q)$. First differentiate Q with respect to P:

$$dQ/dP = 0.2P$$

Then form the elasticity expression using the original Q(P):

$$E(P) = 0.2P \times P / (40 + 0.1P^2) = 0.2P^2 / (40 + 0.1P^2)$$

### Mark Allocation — 4(a)(i)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **5** | **Correctly differentiates Q to get dQ/dP = 0.2P AND correctly forms E(P) = (dQ/dP)(P/Q) with the full expression for Q in the denominator.** |
| Partial | 3 | Correct dQ/dP but elasticity formula assembled incorrectly (e.g. Q/P used instead of P/Q). |
| Minimal | 0 | Incorrect differentiation of Q with respect to P. |

## 4(a)(ii) [5 marks]

*Find the price elasticity of supply when the current price is $P = 13$.*

### FULL MODEL ANSWER

Substitute $P = 13$ into both $dQ/dP$ and $Q$:

$$dQ/dP = 0.2(13) = 2.6$$

$$Q = 40 + 0.1(13)^2 = 40 + 16.9 = 56.9$$

$$E(13) = 2.6 \times (13 / 56.9)$$

$$E(13) \approx 0.594$$

**Checknote:** *$E(13) < 1$ means supply is inelastic at this price — a useful check against part (iii), since it means the percentage change in Q should come out smaller than the percentage change in P.*

### Mark Allocation — 4(a)(ii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **5** | **Correctly substitutes $P = 13$ into both $dQ/dP$ and $Q(P)$ AND reaches $E(13) \approx 0.594$, carried through consistently from the student's own (a)(i) expression.** |
| Good | 3 | Correct substitution into $dQ/dP$ and $Q$ with an arithmetic slip in the final ratio. |
| Partial | 1 | Only one of $dQ/dP$ or $Q$ correctly evaluated at $P = 13$. |
| Minimal | 0 | No substitution attempted. |

### 4(a)(iii) [2 marks]

*What is the percentage change in supply if the price rises by 2%?*

#### FULL MODEL ANSWER

By definition, $E \approx (\%\Delta Q) / (\%\Delta P)$, so $\%\Delta Q = E \times \%\Delta P$:

$$\%\Delta Q = 0.594 \times 2\%$$

$$\%\Delta Q \approx 1.19\%$$

#### Mark Allocation — 4(a)(iii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **2** | **Correctly applies $\%\Delta Q = E \times \%\Delta P$ using the student's own E(13), reaching ≈1.19%.** |
| Minimal | 0 | Formula not applied, or elasticity from (a)(ii) not correctly carried forward. |

## 4(b)(i) [10 marks]

*A company sells Q items at a cost of P per item. The demand function for the items is P = 200 − 4Q. The fixed costs function is FC(Q) = Q² + 50Q + 10. Find the value of Q that maximises the profit and the corresponding price.*

### FULL MODEL ANSWER

**Revenue:** R = P × Q, using the demand function to express price in terms of Q:

$$R(Q) = (200 - 4Q)Q = 200Q - 4Q^2$$

**Profit** = Revenue − Costs. Note the "fixed costs" function here is itself a function of Q — the full cost function — so:

$$\pi(Q) = R(Q) - FC(Q) = (200Q - 4Q^2) - (Q^2 + 50Q + 10)$$

$$\pi(Q) = -5Q^2 + 150Q - 10$$

Maximise by setting the first derivative to zero:

$$d\pi/dQ = -10Q + 150 = 0 \;\Rightarrow\; Q = 15$$

Check the second derivative: $d^2\pi/dQ^2 = -10 < 0$, confirming a **maximum**.

**CONFIRMED MAXIMUM**

Corresponding price, from the demand function:

$$P = 200 - 4(15) = 200 - 60$$

$$Q^* = 15 \text{ units},\;\; P^* = €140$$

**Checknote:** *The label 'fixed costs function' is a deliberate misnomer in this question — FC(Q) here still depends on Q, so it must be subtracted as a full function, not treated as a constant the way a genuinely fixed cost would be.*

### Mark Allocation — 4(b)(i)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **10** | **Correctly forms revenue from the demand function AND correctly forms profit as R − FC AND sets dπ/dQ = 0 to get Q = 15 AND verifies it is a maximum via the second derivative AND correctly finds P = €140.** |
| Good | 7 | Correct profit function and correct Q = 15, but the second-derivative maximum check or the final price is missing. |
| Partial | 4 | Correct revenue function formed, but FC(Q) subtracted incorrectly (e.g. treated as a constant rather than a function of Q). |
| Minimal | 1 | Profit not correctly formed as revenue minus cost. |

## 4(b)(ii) [3 marks]

*Find the total profit for this level of production.*

### FULL MODEL ANSWER

Substitute Q = 15 into the profit function derived in (b)(i):

$$\pi(15) = -5(15)^2 + 150(15) - 10 = -1\{,\}125 + 2\{,\}250 - 10$$

$$\pi(15) = €1{,}115$$

### Mark Allocation — 4(b)(ii)

| Tier | Marks | Criteria |
|---|---|---|
| **Full credit** | **3** | **Correctly substitutes Q = 15 into the profit function from (b)(i) AND reaches €1,115, carried through consistently from the student's own (b)(i) profit function.** |
| Partial | 1 | Correct substitution method but an arithmetic slip in the final value. |
| Minimal | 0 | Incorrect or missing substitution. |

**Question 4 — Total** 25/25 marks confirmed